ECON 695 Problem Set 3 Solutions

October 4, 2025

1. Suppose you have a random sample $\{(x_i, y_i)\}$ of size N, where each y_i is 1×1 (a scalar outcome), and each x_i is also 1×1 (a scalar regressor). If you run a linear regression of y_i on x_i without a constant, show the regression coefficient (OLS estimator) is:

$$\hat{\beta} = \frac{\frac{1}{N}\sum_i y_i x_i}{\frac{1}{N}\sum_i x_i^2} \tag{1}$$

Hint: $\hat{\beta} = argmin_b \frac{1}{N}\sum_i (y_i - x_i b)^2$.

- **Solution:** The first order condition of the hint:

$$\frac{\partial \frac{1}{N}\sum_i (y_i - x_i b)^2}{\partial b} = -2\frac{1}{N}\sum_i x_i(y_i - x_i b) = 0$$

$$\Rightarrow \hat{\beta} = \frac{\frac{1}{N}\sum_i x_i y_i}{\frac{1}{N}\sum_i x_i^2}$$

Note without controlling for a constant, the numerator is $\frac{1}{N}\sum_i x_i y_i$, not the sample covariance: $c\hat{o}v(x_i, y_i) = \frac{1}{N}\sum_i x_i y_i - \bar{x}\bar{y}$ unless $\bar{x}\bar{y} = 0$.

2. Prove the Frisch-Waugh theorem for the sample OLS regression coefficients by showing that the j^{th} row of $\hat{\beta}$ is:

$$\hat{\beta}_j = [\frac{1}{N}\sum_{i=1}^{N} \hat{\xi}_i^2]^{-1}[\frac{1}{N}\sum_{i=1}^{N} \hat{\xi}_i y_i] \tag{2}$$

where $\hat{\xi}_i$ is the *estimated residual* from an auxiliary OLS regression of x_{ji} on all the other $x's$ (denoted by $x_{(-j)i}$):

$$x_{ji} = x'_{(\sim j)i}\hat{\pi} + \hat{\xi}_i. \tag{3}$$

HINT: review lecture 4.

- **Solution:** From the auxiliary regression (3), we have the residual (after partialing out $x_{(\sim j)i}$) is orthogonal to the covariates $x_{(\sim j)i}$:

$$\hat{\xi}_i \perp x_{(\sim j)i} \leftrightarrow \frac{1}{N}\sum_i \hat{\xi}_i x_{(-j)i} = \mathbf{0} \text{ a vector of zeros.}$$

Given the OLS regression $y_i = \hat{\beta}_j x_j + \sum_{k \neq j} \hat{\beta}_k x_k + \hat{u}_i$ where $\hat{\beta} = argmin_b \frac{1}{N}\sum_i (y_i - x_i' b)^2$, we have the residual $\hat{u}_i \perp x_{ki}$ or $\frac{1}{N}\sum_i \hat{u}_i x_{ki} = 0$ for every column of x. We plug y_i into the numerator of (2):

$$\frac{1}{N}\sum_i \hat{\xi}_i y_i = \frac{1}{N}\sum_i \hat{\xi}_i(\hat{\beta}_j x_j + \sum_{k\neq j}\hat{\beta}_k x_k + \hat{u}_i)$$
$$= \frac{1}{N}\left(\hat{\beta}_j \sum_i \hat{\xi}_i x_{ji} + \sum_{k\neq j}\hat{\beta}_k \underbrace{\sum_i \hat{\xi}_i x_{ki}}_{=0} + \sum_i \hat{\xi}_i \hat{u}_i\right)$$

in which $\sum_i \hat{\xi}_i x_k = 0$ for any $k \neq j$

$$\sum_i \hat{\xi}_i \hat{u}_i = \sum_i (x_{ji} - x'_{(\sim j)i}\hat{\pi})\hat{u}_i = 0$$
$$\sum_i \hat{\xi}_i x_{ji} = \sum_i \hat{\xi}_i(x'_{(\sim j)i}\hat{\pi} + \hat{\xi}_i) = \sum_i \hat{\xi}_i^2$$

Using the orthogonality conditions in the original OLS regression and in the auxiliary regressions, we have shown that $\frac{1}{N}\sum_i \hat{\xi}_i y_i = \hat{\beta}_j \times \frac{1}{N}\sum_i \hat{\xi}_i^2$. That is, $\hat{\beta}_j = (\frac{1}{N}\sum_i \hat{\xi}_i^2)^{-1}(\frac{1}{N}\sum_i \hat{\xi}_i y_i)$.

3. We observe an outcome variable y_i and covariates x_i and z_i in a sample of N observations. Assume that the sample means of x_i and z_i are both 0 (i.e., $\overline{x} = 0,\ \overline{z} = 0$). Consider the regression model:

$$y_i = a_0 + a_1 x_i + v_i \tag{4}$$

which we estimate by OLS, obtaining OLS regression coefficients $\widehat{a}_0$ and $\widehat{a}_1$. Show that if x_i is *orthogonal* to z_i in the observed sample ($\frac{1}{N}\sum_i x_i z_i = 0$), then when you estimate the regression model:

$$y_i = b_0 + b_1 x_i + b_2 z_i + e_i \tag{5}$$

by OLS, you will get

$$\widehat{b}_0 = \widehat{a}_0, \text{ and } \widehat{b}_1 = \widehat{a}_1$$

HINT: Write out the three FOC's for the OLS coefficients from model (5). Compare these with the two FOC's from model (4) and use the information you are given.

- **Solution:** Model (4) is the "short" regression where we drop z_i from the "long" regression (5). Whether the sample estimates change depend on the correlations between the omitted variable and the remaining covariates. The OLS estimates of (4) satisfy:

$$(\hat{a}_0, \hat{a}_1) = argmin_{a_0,a_1}\frac{1}{N}\sum_{i=1}^{N}(y_i - a_0 - a_1 x_i)^2$$
$$\frac{\partial}{\partial a_0} = 0 \Rightarrow \hat{a}_0 = \bar{y} - \hat{a}_1\bar{x} = \bar{y} \text{ as } \bar{x} = 0$$
$$\frac{\partial}{\partial a_1} = 0 \Rightarrow \hat{a}_1 = \frac{\hat{cov}(y_i, x_i)}{\hat{var}(x_i)}.$$

The OLS estimates of the "long" regression (5) satisfy:

$$(\hat{b}_0, \hat{b}_1) = argmin_{a_0,a_1}\frac{1}{N}\sum_{i=1}^{N}(y_i - b_0 - b_1 x_i - b_2 z_i)^2$$
$$\frac{\partial}{\partial b_0} = 0 \Rightarrow \hat{b}_0 = \bar{y} - \hat{b}_1\bar{x} - \hat{b}_2\bar{z} = \bar{y} \text{ as } \bar{x} = \bar{z} = 0$$

We can also find $\hat{b}_1$ by using the Frisch-Waugh Theorem. Consider an auxiliary regression of x_i on $(1, z_i)$:

$$x_i = \hat{\pi}_0 + z_i\hat{\pi}_1 + \hat{\xi}_i$$

which is a univariate regression, and we have $\hat{\pi}_1 = \frac{\hat{cov}(x_i,z_i)}{\hat{var}(z_i)} = 0$ as x_i and z_i are uncorrelated (note correlation $\rho_{xz} := \frac{cov(x,z)}{sd(x)\times sd(z)}$). And $\hat{\pi}_0 = \bar{x} - \bar{z}\hat{\pi}_1 = \bar{x}$. Therefore $\hat{\xi}_i = x_i$. By Frisch Waugh,

$$\hat{b}_1 = \frac{\frac{1}{N}\sum_i y_i\hat{\xi}_i}{\frac{1}{N}\sum_i \hat{\xi}_i^2} = \frac{\frac{1}{N}\sum_i y_i(x_i-\bar{x})}{\frac{1}{N}\sum_i (x_i-\bar{x})^2} = \frac{\hat{cov}(x_i,y_i)}{\hat{var}(x_i)} = \hat{a}_1.$$

We have shown $\hat{a}_0 = \hat{b}_0 = \bar{y}$, and $\hat{b}_1 = \hat{a}_1$.

4. Suppose you can observe the wages, education (yrs of schooling), immigrant status, and race and ethnicity of a random sample of workers.

   (a) Consider two linear regressions of $y_i = \log \text{wage}_i$ on observable characteristics:

   $$\begin{aligned} y_i &= \lambda_0 + \lambda_1 Imm_i + u_i \text{ (short model)} \\ y_i &= \beta_0 + \beta_1 Imm_i + \beta_2 Educ_i + \epsilon_i \text{ (long model)} \end{aligned}$$

   If you estimate the regressions in the random sample, how does the OLS estimator $\hat{\lambda}_1$relate to $\hat{\beta}_1$? Which assumption do you need to have $\hat{\lambda}_1 > \hat{\beta}_1$?
   Hint: use the omitted variable bias, and define the terms in the formula carefully.

   - **Solution:** Consider an auxiliary regression in sample

     $$\begin{aligned} Educ_i &= \hat{\pi}_0 + \hat{\pi}_1 Imm_i + \hat{\xi}_i \\ \frac{1}{N}\sum_i \hat{\xi}_i &= \frac{1}{N}\sum_i \hat{\xi}_i Imm_i = 0 \end{aligned}$$

     Plug the auxiliary regression into the long regression in sample:

     $$\begin{aligned} y_i &= \hat{\beta}_0 + \hat{\beta}_1 Imm_i + \hat{\beta}_2 Educ_i + \hat{\epsilon}_i \\ &= \hat{\beta}_0 + \hat{\beta}_1 Imm_i + \hat{\beta}_2\left(\hat{\pi}_0 + \hat{\pi}_1 Imm_i + \hat{\xi}_i\right) + \hat{\epsilon}_i \\ &= (\hat{\beta}_0 + \hat{\beta}_2\hat{\pi}_0) + \left(\hat{\beta}_1 + \hat{\beta}_2\hat{\pi}_1\right) Imm_i + \left(\hat{\beta}_2\hat{\xi}_i + \hat{\epsilon}_i\right) \;\; \mathbf{(*)} \end{aligned}$$

     Since $\hat{\xi}_i$ is orthogonal to $(1, Imm_i)$ and $\hat{\epsilon}_i$ is orthogonal to $(1, Imm_i, Educ_i)$, we have

     $$\begin{aligned} \frac{1}{N}\sum_i (\hat{\beta}_0 + \hat{\beta}_2\hat{\pi}_0) \times \left(\hat{\beta}_2\hat{\xi}_i + \hat{\epsilon}_i\right) &= 0 \\ \frac{1}{N}\sum_i \left(\hat{\beta}_1 + \hat{\beta}_2\hat{\pi}_1\right) Imm_i \times \left(\hat{\beta}_2\hat{\xi}_i + \hat{\epsilon}_i\right) &= 0 \end{aligned}$$

     which suggest $\hat{\lambda}_0 = \hat{\beta}_0 + \hat{\beta}_2\hat{\pi}_0$, $\hat{\lambda}_1 = \hat{\beta}_1 + \hat{\beta}_2\hat{\pi}_1$ solve the first-order conditions from the short regression:

     $$\begin{aligned} (\hat{\lambda}_0, \hat{\lambda}_1) &= argmin_{a,b}\frac{1}{N}\sum_i (y_i - a - bImm_i)^2 \\ \frac{\partial}{\partial a} &= 0 \rightarrow \frac{1}{N}\sum_i (y_i - \hat{\lambda}_0 - \hat{\lambda}_1 Imm_i) = 0 \\ \frac{\partial}{\partial b} &= 0 \rightarrow \frac{1}{N}\sum_i (y_i - \hat{\lambda}_0 - \hat{\lambda}_1 Imm_i) \times Imm_i = 0 \end{aligned}$$

     Therefore, we have $\hat{\lambda}_1 = \hat{\beta}_1 + \hat{\beta}_2\hat{\pi}_1$, where the omitted variable bias is $\hat{\beta}_2\hat{\pi}_1$. The bias is positive if $\hat{\beta}_2\hat{\pi}_1 > 0$ (for example, education is positively related to earnings, and immigrants are more educated than natives).

(b) Split the immigrants into 3 mutually exclusive groups:

i. Asian immigrants: workers who answer asian = 1, hispanic =0, immigrant =1;

ii. Hispanic immigrants: workers who answer hispanic=1, immigrant=1;

iii. Other immigrants: workers who answer asian=0, hispanic=0, immigrant=1.

Specify a single regression that allows you to estimate the mean log wage of each immigrant group relative to natives (immigrant=0).

- **Solution:** Define 3 indicators the immigrant groups: $D_{1i} = 1$ if i is Asian, $D_{2i} = 1$ if i is Hispanic, and $D_{3i} = 1$ if i is Other immigrant. Natives would have $D_{1i} = D_{2i} = D_{3i} = 0$. Consider a regression of log wage on $(1, D_{1i}, D_{2i}, D_{3i})$:

$$\begin{aligned} y_i &= \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + u_i \\ \text{where } & u_i \perp (1, D_{1i}, D_{2i}, D_{3i}) \end{aligned}$$

As we have shown in regressions with dummies, given the four mutually exclusive groups (natives +3 groups of immigrants),

$$\begin{aligned} \beta_0 &= E[y_i|\text{Natives}] \\ \beta_1 &= E[y_i|D_{1i} = 1] - E[y_i|\text{Natives}] \\ \beta_2 &= E[y_i|\text{Hispanic}] - E[y_i|\text{Natives}] \\ \beta_3 &= E[y_i|\text{Other Imm}] - E[y_i|\text{Natives}]. \end{aligned}$$

Note in this question you don't prove the equations above. It's sufficient to specify the regression that will get you the correct answers. In general, you need to prove $\beta_0 = E[y_i|Natives]$ *and* $\beta_k = E[y_i|D_{ki} = 1] - E[y_i|Natives]$.

(c) How much of the (log) wage gaps between each immigrant group and native workers can be explained by differences in education attainment? Specify the regression(s) you will need to estimate to answer this question, and provide an answer from your regressions.
Hint: consider a pooled regression with education as a control, or separate regressions by immigrant group. Focus on the "between" component of the decomposition.

- **Solution:** It is sufficient to consider a single immigrant group, and apply the solution to the other two groups. Suppose we focus on Asian (non-Hispanic) immigrants. Consider separate regressions of log wage on education and a constant:

$$\begin{aligned} \text{(a) Asian: } y_i &= \beta_0^a + \beta_1^a Educ_i + \epsilon_i \\ \text{(b) Natives: } y_i &= \beta_0^b + \beta_1^b Educ_i + \epsilon_i \end{aligned}$$

Then the difference in mean log wage can be decomposed as follows:

$$\begin{aligned} \bar{y}^a - \bar{y}^b &= (\hat{\beta}_0^a + \hat{\beta}_1^a \overline{Educ}^a) - (\hat{\beta}_0^b + \hat{\beta}_1^b \overline{Educ}^b) \\ &= (\hat{\beta}_0^a - \hat{\beta}_0^b) + \hat{\beta}_1^a \overline{Educ}^a - \hat{\beta}_1^b \overline{Educ}^b \\ &= \underbrace{(\hat{\beta}_0^a - \hat{\beta}_0^b)}_{\text{unexplained by educ}} + \underbrace{\left(\hat{\beta}_1^a - \hat{\beta}_1^b\right) \overline{Educ}^a}_{\text{within}} + \underbrace{\hat{\beta}_1^b \left(\overline{Educ}^a - \overline{Educ}^b\right)}_{\text{between}} \end{aligned}$$

The difference that can be explained by diff in education attainment, i.e. the composition effect, is represented by $\hat{\beta}_1^b \left(\overline{Educ}^a - \overline{Educ}^b\right)$.
Note you may also propose a pooled regression. The key is to point out the between component of the difference given your model.

5. You are tasked to provide a causal analysis of how the inflow of Asian immigrants affects the earnings of native workers. Assume you have a county c-level data $\{y_c, x_c, z_c\}$:

- y_c: the change in real log wage among native workers in county c from 1990 to 2000;

- x_c: the percent change in the number of immigrants in county c from 1990 to 2000 (if N_{ct} denote the number of immigrants in county c in year t, $x_c = 100 * (N_{c,2000} - N_{c,1990})/N_{c,1990}$.
- z_c: the share of restaurants in county c that are owned by Asian immigrants in 1990.

The causal model of interest is:

$$y_c = \beta_1 + \beta_2 x_c + u_c \tag{6}$$

in which β_2 is the causal effect of a 1% increase in immigrant population on the change in log wage of native workers, holding all else constant.

(a) Consider an OLS regression of y_c on x_c,

$$\hat{y}_c = \hat{\beta}_1 + \hat{\beta}_2 x_c \tag{7}$$

how would you interpret the coefficient $\hat{\beta}_2$ on x_c? If $cov(x_c, u_c) = 0$ and $E[u_c] = 0$, show that $\hat{\beta}_2$ is a consistent estimator for β_2 in (6).

- **Solution:** $\hat{\beta}_2$ is the predictive effect of the change in immigrants on the log wage of natives. If $cov(x_c, u_c) = 0$, model (6) represents the population regression of y_c on x_c where the residual defined as $u_c = y_c - \beta_1 - \beta_2 x_c$ is orthogonal to the covariates 1 and x_c. The sample OLS estimator in (7):

$$\begin{aligned}
\hat{\beta}_2 &= \frac{\frac{1}{N}\sum_i (x_c - \bar{x}) y_c}{\frac{1}{N}\sum_i (x_c - \bar{x})^2} = \frac{\frac{1}{N}\sum_i (x_c - \bar{x})(\beta_1 + \beta_2 x_c + u_c)}{\frac{1}{N}\sum_i (x_c - \bar{x})^2} \\
&= \beta_2 + \frac{\frac{1}{N}\sum_i (x_c - \bar{x}) u_c}{\frac{1}{N}\sum_i (x_c - \bar{x})^2} \\
&\xrightarrow{p} \beta_2 + \frac{cov(x_c, u_c)}{var(x_c)} = \beta_2 \text{ given } cov(x_c, u_c) = 0.
\end{aligned}$$

  - Note I forgot to put $E[u_c] = 0$, but even without this condition, we can consider the population regression $y_c = (\beta_1 + E[u_c]) + \beta_2 x_c + \widetilde{u}_c$, where $\widetilde{u}_c = u_c - E[u_c]$ is orthogonal to both 1 and $x_c (E[\widetilde{u}_c] = E[\widetilde{u}_c x_c] = 0)$.

(a) If $E[u_c x_c] \neq 0$ in model (6), you may want to look for an instrument for the endogenous x_c. What assumptions do you need for z_c as defined to be a valid instrument for x_c?

- **Solution:** The instrument(s) z_c for the endogenous variable x_c should satisfy two conditions:

$$\begin{aligned}
cov(x_c, z_c) &\neq 0 \text{ relevance} \\
E[z_c u_c] &= 0 \text{ exclusion restriction}
\end{aligned}$$

  (1) the instrument is relevant for x_c (as a strong predictor), and (2) the instrument is uncorrelated with the unexplained residual u_c in the causal model – z_c can only affect the outcome through x_c.

(a) Suppose the assumptions for z_c to be an valid instrument for $x_c$are satisfied. Consider the first-stage regression in the population

$$x_c = \pi_0 + \pi_1 z_c + \nu_i$$

and the reduced-form regression (population)

$$y_c = \delta_0 + \delta_1 z_c + \epsilon_i.$$

Show:

$$\frac{\delta_1}{\pi_1} = \beta_2.$$

- **Solution:** Plug the first-stage regression into the causal model:

$$
\begin{aligned}
y_c &= \beta_1 + \beta_2 \left(\pi_0 + \pi_1 z_c + \nu_i\right) + u_c \\
&= (\beta_1 + \beta_2 \pi_0) + (\beta_2 \pi_1)\, z_c + (\beta_2 \nu_i + u_c)
\end{aligned}
$$

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$\Pi(1, z_c) \Rightarrow \delta_0 = \beta_1 + \beta_2 \pi_0$

$\delta_1 = \beta_2 \pi_1$

In the first stage regression, the residual satisfies: $E[\nu_i] = E[\nu_i z_c] = 0$. By exclusion restriction we also have $E[z_c u_c] = 0$. Therefore in the model above we have

$$
\begin{aligned}
E[z_c(\beta_2 \nu_i + u_c)] &= 0 \\
E[(\beta_2 \nu_i + u_c)] &= 0
\end{aligned}
$$

which correspond to the first-order conditions for the reduced form model:

$$
(\delta_0, \delta_1) = argmin_{a,b} E[(y_c - a - b z_c)^2].
$$

The solution to the problem above is unique (z_c is not a constant), and we have

$$
\delta_1 = \beta_2 \pi_1.
$$

(a) Do you think z_c as defined is a good instrument for x_c? Why or why not? Hint: this is an open-ended question. You can argue either way.

- **Solution:** The first stage should work, but you may argue the exclusion restriction may or may not hold.
  - (1) z_c is not a good instrument since it's ~~likely~~ unlikely to satisfy the exclusion restriction. A higher share of restaurants owned by Asian immigrants may be correlated with economic growth that matters for the earnings of natives, resulting in $E[z_c u_c] \neq 0$.
  - (2) z_c may be a good instrument if earlier waves of Asian immigrants settled in a city for exogenous reasons that are uncorrelated with current economic conditions.